University of Toledo

Boot Sense: A Sensor-Based Orthopedic Boot for Real-Time Partial Weight-Bearing Monitoring

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Abstract

Boot Sense is a low-cost, sensor-based system designed to help patients recovering from lower-limb injuries safely adhere to partial weight-bearing restrictions while using a walking boot. Patients are frequently instructed to limit load to a percentage of their body weight, yet current clinical tools are expensive, inaccessible, or provide no real-time feedback outside the clinic. Boot Sense addresses this gap by integrating two strain-gauge load cells, an HX711 instrumentation amplifier, and an Arduino MKR WiFi 1010 microcontroller to measure applied load and provide immediate vibration and LED alerts when prescribed limits are exceeded.

Engineering principles in biomechanics, strain-gauge mechanics, signal conditioning, and embedded control guided the design. A calibration process based on empty-boot and full-weight readings allows the system to approximate percent body weight, while a hysteresis-based algorithm (≥30% activation, ≤20% deactivation, 300 ms hold time) ensures stable feedback without noise-triggered alerts. Prototype testing demonstrated that the system reliably detects load changes, produces predictable ADC behavior after averaging, and activates alerts consistently under static loading conditions.

Testing also identified important limitations: the two load cells are combined through a single amplifier channel, preventing independent heel and forefoot sensing; electronics remain external and non-wearable; and power delivery through portable battery packs caused voltage instability. These findings highlight necessary improvements including dual-channel sensing, embedded electronics, a regulated onboard battery, and a clinician-adjustable interface for setting thresholds.

Despite these limitations, Boot Sense successfully validates the feasibility of a wearable rehabilitation aid capable of providing real-time feedback during partial weight-bearing. With further refinement, the system has strong potential to enhance patient compliance, reduce reinjury risk, and expand access to intelligent orthopedic care.

Introduction

Lower-limb injuries often require strict non-weight-bearing or partial-weight-bearing restrictions during recovery, yet compliance with these instructions is notoriously poor. According to NIH data, 62.5% of patients prescribed weight-bearing limitations unknowingly exceed their allowed load, which can lead to delayed healing or reinjury [33]. This highlights a significant gap in orthopedic rehabilitation: traditional walking boots provide structural support but no real-time feedback to help patients understand how much force they are placing on their injured limb. Boot Sense directly addresses this unmet clinical need by integrating load-monitoring sensors and real-time alerts into a standard medical walking boot to improve patient compliance and enhance recovery outcomes.

Boot Sense uses two load cells embedded beneath the foot to continuously measure the force applied during standing and walking. During setup, the system records an "empty" baseline and a patient-specific "stand" value representing full body weight. These calibration values allow the device to determine whether the patient is loading their injury within safe limits prescribed by their physician. Boot Sense implements a hysteresis-based threshold structure to ensure stable performance: the alert activates when the patient reaches or exceeds 30% of their calibrated standing weight for more than 300 milliseconds, and the alert turns off only when the load decreases below 20%. This prevents rapid switching of the alert when the load fluctuates near the threshold. When excessive load is detected, the system triggers a vibration motor and LED indicator to cue the patient to offload weight. The device is powered by a rechargeable battery, and although threshold adjustments are currently made through the device's embedded code, future versions may include a keypad, touch interface, or computer-based configuration tool to allow clinicians to easily adjust loading limits for each patient.

In the broader context of biomedical engineering, Boot Sense reflects the ongoing shift toward intelligent rehabilitation devices that provide real-time biomechanical insight to improve patient outcomes. Patients often cannot accurately sense whether they are placing too much force on an injured limb, especially in the absence of pain, leading to accidental overload and delayed healing. By translating sensor data into intuitive alerts, Boot Sense supports safer movement patterns, promotes adherence to medical instructions, and reduces the likelihood of reinjury or complications that increase healthcare costs.

Boot Sense enters a rapidly growing market at the intersection of orthopedic rehabilitation devices and wearable medical technology. According to global health estimates, ankle injuries alone affect over 25 million people worldwide each year, representing a substantial population that could benefit from safer and more guided rehabilitation tools [34]. Despite this large patient population, none of the standard walking boots available today integrate real-time load monitoring, threshold-based alerts, or smart rehabilitation features. This gives Boot Sense a unique competitive advantage within a multibillion-dollar orthopedic device market. As healthcare providers increasingly adopt outcome-based care and insurers emphasize prevention of avoidable complications, a device that improves patient compliance and reduces reinjury risk holds significant commercial potential. Boot Sense aligns with broader industry trends, including smart wearables, remote monitoring, and personalized rehabilitation, positioning it as a scalable and impactful solution in modern orthopedic care. Projected early-market adoption and revenue potential are summarized in Table 1.

Table 1. Projected unit sales and revenue for Boot Sense during the first three years of commercialization.

Year	Units Sold	Revenue
Year 1	5,000 – 8000	$1.25M – $2M
Year 2	12,000 – 18,000	$3M – $4.5M
Year 3	20,000 – 30,000	$5M – $7M

By combining reliable sensing hardware, intuitive feedback mechanisms, and evidence-based design strategies, Boot Sense offers a meaningful advancement over traditional passive walking boots. Its foundation in biomedical engineering principles, its alignment with current healthcare needs, and its practical implementation in patient rehabilitation position Boot Sense as a promising, clinically useful device with strong potential for real-world impact.

Team Organization

The Boot Sense team consists of five senior bioengineering students: Bailee, Raquel, Hailey, Ilina, and Shereen. At the beginning of the academic year, the project operated under a shared leadership model in which each member participated in planning, brainstorming, and initial prototyping. As the device advanced into more technically complex stages, responsibilities naturally shifted toward areas of individual strength.

As the project progressed, Shereen took on an expanded technical role integrating the sensing system, refining the clinical rationale for weight-bearing thresholds, implementing the hysteresis-based alert algorithm, and preparing most of the final documentation. Her background in bioengineering design, clinical experience, and programming enabled her to guide much of the system-level decision-making and ensure alignment with patient and engineering best practices.

Other team members contributed in ways that supported early-stage development. Bailee assisted with initial prototyping logistics and preparing presentation materials. Raquel contributed to scheduling and early-stage business. Hailey provided support related to budgeting and component ordering. Ilina assisted with parts research and regulatory background information during the design process.

Collaboration was maintained through regular team meetings, shared OneDrive folders for documentation, and ongoing communication to keep members aligned on deadlines and design updates. While the distribution of work evolved over the project timeline, this structure allowed all members to contribute within their abilities while ensuring that the technical aspects were completed to meet course requirements.

The University of Toledo provided essential support through access to laboratory space, tools, and prototyping. Components were primarily sourced through Amazon due to cost-effectiveness and rapid delivery. The team also received guidance from two faculty advisors: Dr. Brent Cameron, who offered expertise related to sensors and instrumentation, and Dr. Yuan Tang, the Senior Design instructor who oversaw project expectations and documentation.

Despite the challenges inherent in coordinating a large undergraduate team, the group successfully brought Boot Sense from concept to functional prototype. The final system represents a combination of shared planning efforts and concentrated technical development necessary to meet the project deliverables.

Design Methodology

a. Approaches, thought processes, and engineering principles

The design of Boot Sense began with the clinical problem: patients prescribed non-weight-bearing (NWB) or partial-weight-bearing (PWB) instructions lack any intuitive or affordable way to monitor real-time loading on their injured limb [34, 35, 37]. Studies show that over half of patients unintentionally exceed their prescribed limits during recovery, contributing to delayed healing and reinjury [34, 35, 37]. This motivated the development of a device that measures load and provides an immediate alert when unsafe thresholds are exceeded.

From this problem statement, the team defined core engineering requirements:

- **Accurate load measurement** from inexpensive sensors
- **Stable signal conditioning** suitable for home use
- **Reliable, intuitive feedback** via vibration and LED
- **Low power consumption** for daily wear
- **Simple calibration** adaptable to different users and boot types

Meeting these requirements relied on principles from multiple engineering domains:

- **Biomechanics:** Guided selection of clinically meaningful thresholds for partial weight-bearing (~20–50% of body weight) [34, 35, 37].
- **Mechanics and strain-gauge theory:** Informed use of load cells that convert shell deformation into electrical signals [36, 38].
- **Electrical engineering**: Directed integration of the HX711 instrumentation amplifier to convert microvolt-level Wheatstone bridge outputs into stable digital readings [36, 38].
- **Mathematics:** Supported normalization of raw ADC values to percent body weight using two calibration points (EMPTY and STAND).

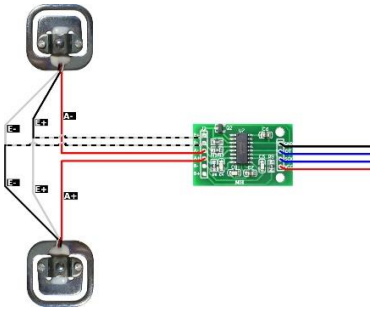

Figure 1. Diagram showing the connection of load cells, HX711 amplifier, and Arduino MKR WiFi 1010 microcontroller for real-time load acquisition and control of vibration and LED.

Feedback output was provided through a vibration motor and a supporting LED indicator, which mirrored each other to offer both tactile and visual alert.

b. Engineering analyses and design iterations

Boot Sense progressed through several design iterations informed by electrical, mechanical, and computational testing.

Initial Prototype

The earliest prototype followed a linear chain:

load cell → HX711 → microcontroller → vibrator motor.

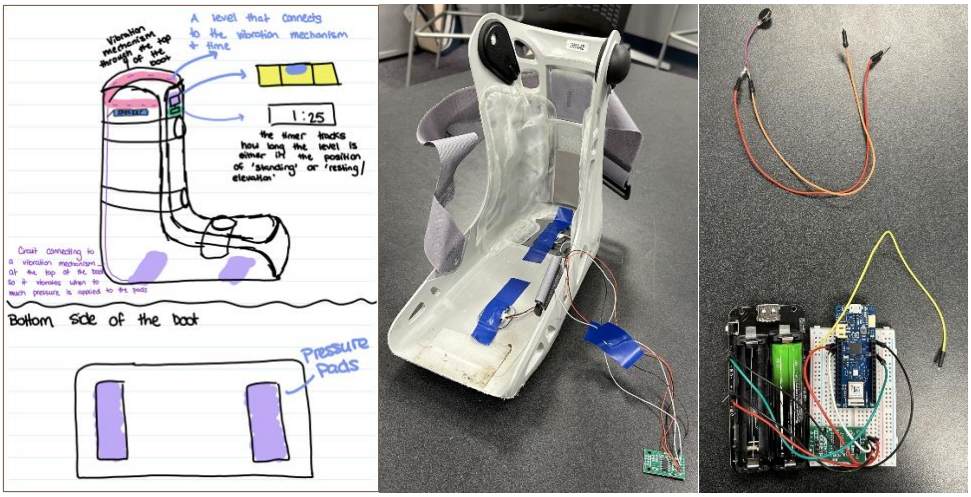

Figure 2. Early prototype development showing (left) the initial sketch of sensor placement, (middle) early boot-mounted load cell configuration, and (right) early wiring of the HX711 + Arduino system used to test initial load detection.

Initial testing revealed challenges:

- Significant noise and unstable raw readings
- A faulty HX711 module causing intermittent communication
- Uneven force transfer between the boot shell and load cells

Mechanical Iterations

Mechanical Iterations

Load cells were repositioned underneath the boot frame to improve load transfer and reduce lateral forces that introduced variability. This substantially improved reading consistency.

Software Refinement

Analysis of raw ADC outputs identified fluctuations of strain-gauge sensors. To stabilize readings, we implemented:

- **Sample averaging** to smooth noise

- **300 ms hold time** so vibration triggers only when sustained thresholds are exceeded
- **Hysteresis logic:**
    - ON at ≥30% BW
    - OFF at ≤20% BW

This prevented rapid buzzing ("flicker") when readings hovered near the threshold. We also integrated an LED indicator in parallel with the vibration motor so that visual confirmation of alert activation could be observed during debugging and testing.

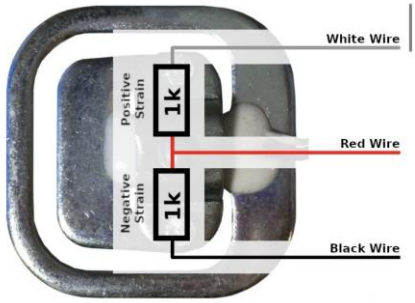

Figure 3. Internal strain-gauge structure of a typical load cell showing the Wheatstone bridge configuration used to convert mechanical deformation into proportional electrical signals.

c. Experimental planning, data collection, and analysis

Experiments were planned to evaluate three major questions:

1. Does the system reliably detect changes in load magnitude?
2. Does the calibration method map raw values to percent body weight accurately?
3. Does the alert trigger at the correct times and remain off otherwise?

Testing consisted of sequential load applications:

- EMPTY boot resting on the ground
- Partial loading using controlled amounts of pressure
- Full standing weight

For each condition, raw HX711 values were recorded using the Arduino Serial Monitor. Multiple samples were collected per condition to assess repeatability.

Calibration Method

We used two calibration points:

- EMPTY = raw ADC reading with no load
- STAND = raw ADC reading with full body weight

$$\%BW = \frac{reading - EMPTY}{STAND - EMPTY}$$

This creates a **linear normalization** from 0 to 100% of body weight without requiring any external calibration weights.

Verification

The device was tested to confirm:

- % BW increased appropriately with load
- Vibration activated when % BW ≥ 30% for ≥ 300 ms
- Vibration deactivated when %BW below 20%
- Hysteresis prevented flicker

The final prototype demonstrated stable readings and alert behavior under realistic conditions.

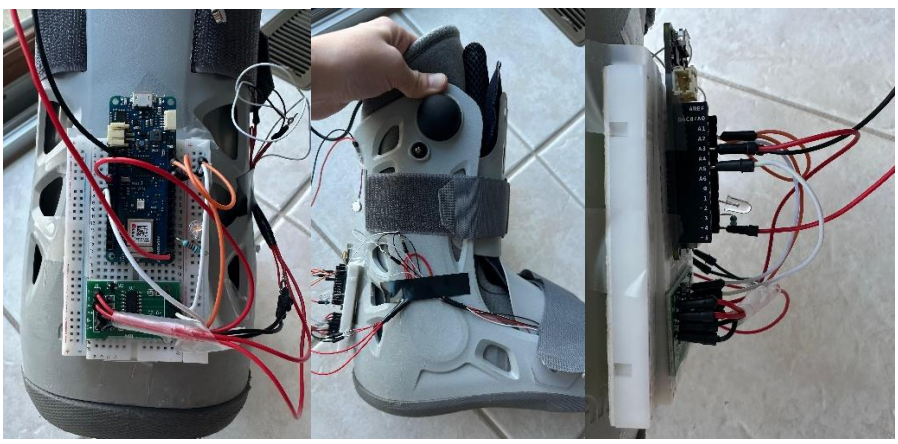

Figure 4. Final prototype hardware setup showing sensor placement under the boot frame and final wiring mounted to the boot shell.

```
13:39:14.589 -> raw = -666790   %BW = 0.00
13:39:15.434 -> raw = -667014   %BW = 0.00
13:39:16.298 -> raw = -667194   %BW = 0.00
13:39:17.084 -> raw = -667256   %BW = 0.00
13:39:17.915 -> raw = -667301   %BW = 0.00
13:39:18.768 -> raw = -666608   %BW = 0.00
13:39:19.586 -> raw = -663723   %BW = 0.91
13:39:20.449 -> raw = -631043   %BW = 24.26
13:39:21.261 -> raw = -453118   %BW = 150.00
13:39:22.097 -> raw = -417877   %BW = 150.00
13:39:22.905 -> raw = -428304   %BW = 150.00
13:39:23.775 -> raw = -418398   %BW = 150.00
13:39:24.601 -> raw = -421476   %BW = 150.00
13:39:25.426 -> raw = -414974   %BW = 150.00
13:39:26.246 -> raw = -416512   %BW = 150.00
13:39:27.079 -> raw = -415001   %BW = 150.00
13:39:27.914 -> raw = -428626   %BW = 150.00
13:39:28.726 -> raw = -616273   %BW = 34.80
13:39:29.576 -> raw = -664692   %BW = 0.22
13:39:30.389 -> raw = -665482   %BW = 0.00
13:39:31.259 -> raw = -667934   %BW = 0.00
13:39:32.079 -> raw = -669356   %BW = 0.00
```

Figure 5. Serial Monitor output demonstrating stable raw readings and accurate %BW conversion during testing.

(a)

```
// Simple averaging to reduce noise
long readAverage(int n) {
  long sum = 0;
  for (int i = 0; i < n; i++) {
    while (!scale.is_ready()) { /* wait for
conversion */ }
    sum += scale.read();
  }
  return sum / n;
}
```

(b)

```
void loop() {
  // 1) Read sensor and average a few samples
  long reading = readAverage(10);

  // 2) Convert to % body weight on that foot: 0.0 = no load, 1.0 =
full BW
  float pctBW = 0.0;
  if (DELTA != 0) {
    pctBW = (float)(reading - EMPTY) / (float)DELTA;
  }
```

(c)

```
// 3) Clinical-style logic: threshold + hold time + hysteresis

  if (!vibOn) {
    // Currently OFF → check if we should turn ON
    if (pctBW >= PCT_ON) {   // above ~30% BW
      if (aboveStart == 0) {
        aboveStart = now;    // start timing sustained load
      } else if (now - aboveStart >= HOLD_MS) {
        vibOn = true;        // above threshold long enough -> ON
      }
    } else {
      aboveStart = 0;        // dropped below threshold -> reset
timer
    }
  } else {
    // Currently ON → turn OFF only when below lower threshold
(hysteresis)
    if (pctBW <= PCT_OFF) {  // below ~20% BW
      vibOn = false;
      aboveStart = 0;
    }
  }

  // 4) Apply to vibrator + LED (LED mirrors vibrator for easy
visual feedback)
  digitalWrite(VIB_PIN, vibOn ? HIGH : LOW);
  digitalWrite(LED_PIN, vibOn ? HIGH : LOW);

  delay(50);  // small delay for serial readability
}
```

Figure 6. Code snippets used for averaging (a), %BW conversion (b), and hysteresis-based threshold logic (c).

d. New knowledge, techniques, and tools

Throughout development, the team gained significant hands-on skills:

- Soldering and assembly of sensor-based circuits
- Interfacing strain-gauge load cells in a Wheatstone bridge configuration
- Use of the HX711 amplifier for signal conditioning
- Real-time data acquisition, filtering, and actuator control within Arduino
- Debugging faulty hardware and communication inconsistencies
- Embedding sensors within a dynamic mechanical environment

- Controlling multiple actuator outputs (vibration motor + LED) in parallel for real-time feedback

These skills integrated electrical, mechanical, and computational parts into a unified system.

e. Final engineering design constraints

The final prototype of Boot Sense reflects several interacting design constraints:

- **Clinical:** Must detect medically relevant threshold load levels without affecting gait or boot functionality.
- **Mechanical:** Components must fit into the boot structure without adding bulk and must withstand repeated loading.
- **Electrical:** System must run on a rechargeable battery, deliver safe current to actuators, and maintain wire durability.
- **Cost:** Must stay near the $250 target price, limiting hardware complexity.
- **Usability:** Calibration must remain simple and vibration feedback intuitive.
- **User variability:** Must accommodate differing gaits, body weights, and boot models.

These constraints shaped the final function of Boot Sense and define areas for future refinement.

Issues Related to the Design & Manufacturing of the Device

a. Intellectual property issues

Based on a current prior-art review conducted this semester using Google Patents and USPTO databases, we did not identify any granted utility patents that combine both pressure sensing and elevation-based feedback within an orthopedic CAM boot. While several patents exist for pressure-sensing insoles, none describe an integrated real-time feedback system embedded directly into a post-operative walking boot [20][21].

For Boot Sense, the appropriate intellectual property pathway would begin with filing a provisional patent application (PPA) [17]. This provides "patent pending" status and secures a priority date while allowing us to continue refining the design over the next 12 months. Because a PPA does not require formal drawings or finalized claims, it accommodates the iterative nature of our current prototype development.

After further refinement and validation, a non-provisional utility patent would be filed to define the legal claims and undergo examination by the USPTO [17]. The Boot Sense name may qualify for trademark protection, and aspects of the firmware could be protected as trade secrets.

Although commercial systems such as Orpyx and Moticon offer smart insoles for gait monitoring, none combine pressure and elevation alerts within a single orthopedic boot structure [18][19]. Before commercialization, a full freedom-to-operate (FTO) analysis would be required.

b. Regulatory process and approval issues (FDA)

Boot Sense is intended for use during orthopedic rehabilitation and would likely be regulated as a wearable medical device. Depending on final claims, it would fall under FDA Class I or Class II. Class II devices typically require a 510(k) submission demonstrating substantial equivalence to an existing product [22][23].

To prepare for potential future commercialization, we recognize the need to comply with FDA design controls, quality documentation, potential clinical validation, and manufacturing standards. If future iterations include wireless connectivity, the device may also fall under FCC regulations for electronic communication devices [24]. ISO 13485 guidelines would help establish quality assurance and traceability during scaling [25][26].

Because the current semester prototype provides real-time alerts without storing identifiable patient data, HIPAA considerations are minimal at this stage. If future versions incorporate data logging or app connectivity, privacy and cybersecurity requirements would increase [27].

c. Safety and hazard issues

Safety remained a primary focus throughout this semester's design and prototyping. Electrically, the system operates on low voltage from a battery-powered Arduino, eliminating shock hazards. All wiring was soldered, insulated, and strain-relieved to prevent shorts or wire fatigue inside the boot.

False positives or negatives could compromise user trust or safety. To address this, we implemented signal averaging, hysteresis, and a minimum activation hold time, ensuring feedback only triggers when meaningful load is applied. Mechanically, components were positioned so they did not alter the boot's structural integrity or interfere with gait. Throughout this semester, we evaluated stability, wire routing, sensor compression, and potential discomfort to ensure safe integration.

d. Ethical issues concerning the manufacture or use of the device

Ethical considerations center on patient safety, truthful communication of device capability, and accessibility. Boot Sense is intended as an assistive feedback tool, not a diagnostic device; users must follow clinical instructions. Ethical use requires transparency about limitations, like sensor variability, calibration differences among users, and durability over time.

Affordability is another ethical priority. Many medical devices are priced in ways that limit access, so our low-cost design emphasizes reducing disparities in orthopedic rehabilitation. If future versions incorporate data tracking, ethical guidelines would require informed consent and clear disclosure of data handling practices.

e. Manufacturability issues and quality control processes

This semester's prototype highlighted several manufacturability challenges. Embedding load cells and electronics into a boot that undergoes daily mechanical stress requires robust mounting methods. Variability in strain-gauge sensitivity and mounting pressure can affect calibration accuracy, meaning future mass production would require standardized jigs and calibration fixtures.

Quality control would include:

- Batch testing load cells
- Verifying HX711 amplifier stability
- Inspecting solder joints and insulation
- Confirming vibration motor responsiveness
- Validating calibration consistency

Scaling production would require transitioning from individual wires to a custom PCB and integrating sensors into a modified boot shell. Establishing repeatable assembly procedures and QC checkpoints would be essential for producing reliable units at the target price.

Results of Prototype Testing

Prototype testing of Boot Sense was conducted to evaluate whether the system could (1) detect load changes using the two load cells, (2) correctly trigger alerts when partial weight-bearing thresholds were exceeded, and (3) operate consistently with the implemented hysteresis and timing logic.

1. **Sensor Output Behavior**

Testing showed that both load cells, routed through a single HX711 channel, produced stable raw ADC values after implementing software averaging.

- The empty boot produced a consistent baseline.
- When the user applied full standing pressure, the system reliably recorded a higher, stable ADC value.

Although both load cells were merged into one channel, the combined signal was sufficient to detect meaningful differences between "no load," "partial load," and "full load," confirming feasibility for a proof-of-concept system.

2. **Calibration Results**

Calibration was performed by capturing two values in real time:

- Empty value (boot unloaded)
- Stand value (user standing with full body weight)

These values were used to compute a linear scaling factor that mapped changes in raw ADC counts to approximate percent body weight (%BW).

Testing confirmed that:

- %BW increased predictably as load increased
- Calibration was repeatable if the user stood evenly on the boot
- The equation behaved consistently across multiple trials

However, uneven stance and shifting weight introduced variability, signaling the need for dual-channel amplification in future versions.

3. **Threshold Activation Testing**

Testing demonstrated that the alert logic functioned as designed:

- The vibration motor and LED indicator activated reliably when the calculated %BW reached or exceeded the 30% threshold for ≥300 ms.
- The alert turned off when load fell below ~20%, verifying correct function of hysteresis
- No flickering or false activations occurred with averaging and the hold-time condition

This confirmed that the control algorithm performed reliably under realistic loading.

4. **Power and Hardware Behavior**

Testing revealed several power-related limitations in the prototype:

- The Arduino MKR WiFi 1010 provides both 5 V and 3.3 V pins, but the HX711 amplifier must be powered at 3.3 V. During early testing, the circuit was mistakenly powered using the 5 V pin, which produced unstable ADC readings and may have contributed to one of the HX711 modules becoming non-functional.
- The small lithium battery pack originally selected for the design failed to power the MKR and sensors reliably. This suggests the pack could not supply the necessary current or the correct regulated output.

Future iterations should integrate a dedicated Li-ion battery with an onboard 3.3 V LDO regulator, add a proper motor driver to isolate vibration load, and include a securely mounted internal battery housing optimized for microcontroller-level current demands.

5. **Mechanical Testing**

Because the electronics were mounted externally and not embedded inside the boot shell, full gait testing (walking) was not performed for safety reasons.

However, the following all produced predictable and repeatable results:

- Static standing tests
- Partial-weight application
- Heel/forefoot loading

6. **Overall Achievement of Design Goals**

Met goals:

- Device senses load reliably
- Alerts activate at preset partial weight-bearing thresholds
- Hysteresis + timing successfully eliminate noise-triggered alerts
- Calibration process is functional and repeatable
- LED + vibration provide clear feedback cues

Partially met goals / unmet goals:

- Load cells cannot yet be analyzed independently due to single-channel input
- Electronics are external and not wearable
- Battery system requires redesign
- Clinician cannot adjust thresholds without modifying code

Conclusion of Results

The results demonstrate that the prototype successfully meets the core functional objective: real-time load monitoring with a stable and reliable alert system. At the same time, testing revealed important limitations involving amplification, mechanical integration, power delivery, and clinical usability, which guide the next phase of design improvements.

Discussion

a. Interpretation of results and recommended changes

Prototype testing demonstrated that Boot Sense successfully performs its core function: detecting load changes and providing real-time alerts when unsafe partial weight-bearing limits are exceeded. The hysteresis logic (≥30% activation, ≤20% deactivation, 300 ms delay) prevented flicker and false activations, and calibration using empty and stand values produced predictable, repeatable results.

However, several limitations emerged during testing that inform recommended design changes:

1. **Single-Channel Sensing Limitation**

Both load cells were routed into **one HX711 amplifier channel**, which restricted the system to a combined load measurement. This prevented independent analysis of heel vs forefoot loading and made the device sensitive to uneven stance.

Recommended improvement:

- Transition to **two HX711 modules** (one for each load cell), or
- Use a **multichannel instrumentation amplifier**,

to allow independent sensing and improved gait-phase detection.

2. **Calibration Variability & Lack of Clinician Interface**

Calibration relied on the user standing to capture %BW scaling values. Thresholds (30%, 20%, 300 ms) are currently hard-coded, meaning clinicians must modify firmware to adjust.

Recommended improvement:

Introduce a **clinician-facing interface** such as:

- keypad or small touchpad,
- Bluetooth app,
- or PC configuration utility,

This allows physicians to adjust thresholds without altering code. These values (i.e. 30%, 20%, 300 ms, etc.) should be adjustable and clinically validated in future testing.

3. **Power Supply Issues**

Testing revealed inconsistent performance when using an external battery pack with the Arduino MKR WiFi 1010. This occurred because:

- The MKR WiFi 1010 and the HX711 both operate on **3.3 V**, but the prototype sometimes used the **5 V pin**, leading to unstable amplifier readings and damaging the amplifier.
- The vibration motor drew enough current to cause voltage dips, affecting sensor output.
- Portable USB battery banks regulate voltage for high-current phone charging, not microcontrollers, and can shut off or fluctuate at low loads.

Recommended improvement:

- Integrate a dedicated Li-ion battery with a 3.3 V low-dropout (LDO) regulator, add a proper motor driver, and design a sealed battery compartment within the boot.

4. **Mechanical & Manufacturability Limitations**

   The prototype's wiring, amplifier, and vibration motor were mounted **externally**, which:

- Prevents safe gait trials
- Exposes wiring to damage
- Does not reflect clinically wearable design

Recommended improvement:

- Embed sensors within the sole or a rigid insole plate,
- Integrate electronics into the **boot shell** or a detachable waterproof module
- Route wires through protected channels
- Finalize a manufacturable internal layout

5. **Alert System Improvements**

   While the vibration motor and LED worked reliably, user accessibility can be improved.

Recommended improvement:

- Maintain LED as a secondary cue
- Consider stronger vibration or audio cues for users with neuropathy or impaired sensation

b. What are the next steps in the development of your device or product?

Based on prototype performance, the next steps include:

1. **Independent dual-channel sensing**
   - Upgrade amplification for separate heel and forefoot data
2. **Embedded electronics housing**
   - Design a sealed enclosure integrated into the boot
3. **Dedicated power system**
   - Replace battery pack with rechargeable onboard battery

4. **Clinician-configurable interface**
    - Allow adjustment of thresholds without firmware editing
5. **Durability and gait testing**
    - Perform walking trials once electronics are embedded
6. **Long-term clinical validation**
    - Verify safety, accuracy, and usability in real rehab settings
7. **Movement toward manufacturability**
    - Develop custom PCB to replace breadboard
    - Integrate sensors into boot molding

Together, these improvements will transition Boot Sense from a proof-of-concept prototype to a wearable, reliable, clinically deployable system.

Conclusion

Boot Sense successfully demonstrates the feasibility of a low-cost, sensor-based system that provides real-time alerts when patients exceed prescribed partial weight-bearing limits during rehabilitation. Through iterative refinement of hardware and software, including averaging, hysteresis, timed activation, and vibration/LED alerts, the prototype performed reliably in static loading tests and met its primary functional goal.

Testing also revealed several essential areas for improvement. The use of a single-channel amplifier limits independent load sensing, the calibration procedure requires firmware access, the external placement of electronics restricts real-world use, and the power system needs redesign for stability. These limitations provide a clear roadmap for future development, including dual-channel sensing, embedded electronics, a clinician-friendly interface, and a dedicated rechargeable battery system.

Overall, the Boot Sense project demonstrates a meaningful step toward an intelligent orthopedic boot capable of improving patient compliance, reducing reinjury risk, and supporting safer recovery. With further refinement and manufacturable integration, Boot Sense has strong potential for real-world clinical impact.

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Appendix A: Full Arduino Source Code

```
#include "HX711.h"

// === HX711 PINS (MKR WiFi 1010) ===
const int DOUT_PIN = A5;   // HX711 DOUT -> A5
const int SCK_PIN  = A3;   // HX711 SCK  -> A3

// === OUTPUT PINS ===
const int VIB_PIN  = 6;    // coin vibrator control (goes to transistor/driver)
const int LED_PIN  = 7;    // status LED (through 220Ω to GND)

HX711 scale;

// ==== CALIBRATION VALUES ====
// Replace these with YOUR measured averages:
const long EMPTY  = -685000;   // avg raw with NO weight on boot
const long STAND  = -525000;   // avg raw with FULL body weight on that foot
// ============================

// Derived from calibration:
const long DELTA      = STAND - EMPTY;  // how many counts = 100% body weight

// ==== "CLINICAL" THRESHOLDS ====
// Partial weight-bearing zone (PT-style):
const float PCT_ON    = 0.30;   // 30% body weight -> turn ON (within PWB range
20–50%)
const float PCT_OFF   = 0.20;   // 20% body weight -> OFF (hysteresis)
// Time requirement to avoid noise/transient steps:
const unsigned long HOLD_MS = 300;  // must be above threshold this long (ms)
// ================================

bool vibOn = false;
unsigned long aboveStart = 0;

// Simple averaging to reduce noise
long readAverage(int n) {
  long sum = 0;
  for (int i = 0; i < n; i++) {
    while (!scale.is_ready()) { /* wait for conversion */ }
    sum += scale.read();
  }
  return sum / n;
}
```

```
void setup() {
  Serial.begin(9600);
  Serial.println("HX711 Clinical Logic + LED + Vibrator");
  scale.begin(DOUT_PIN, SCK_PIN);

  pinMode(VIB_PIN, OUTPUT);
  pinMode(LED_PIN, OUTPUT);

  digitalWrite(VIB_PIN, LOW);
  digitalWrite(LED_PIN, LOW);
}

void loop() {
  // 1) Read sensor and average a few samples
  long reading = readAverage(10);

  // 2) Convert to % body weight on that foot: 0.0 = no load, 1.0 = full BW
  float pctBW = 0.0;
  if (DELTA != 0) {
    pctBW = (float)(reading - EMPTY) / (float)DELTA;
  }

  // Bound it a bit for sanity
  if (pctBW < 0)   pctBW = 0;
  if (pctBW > 1.5) pctBW = 1.5;  // allow up to 150% BW

  // Debug output (good for your validation/testing section)
  Serial.print("raw = ");
  Serial.print(reading);
  Serial.print("   %BW = ");
  Serial.println(pctBW * 100.0);

  unsigned long now = millis();

  // 3) Clinical-style logic: threshold + hold time + hysteresis

  if (!vibOn) {
    // Currently OFF → check if we should turn ON
    if (pctBW >= PCT_ON) {   // above ~30% BW
      if (aboveStart == 0) {
        aboveStart = now;    // start timing sustained load
      } else if (now - aboveStart >= HOLD_MS) {
        vibOn = true;        // above threshold long enough -> ON
      }
    } else {
```

```
      aboveStart = 0;        // dropped below threshold -> reset timer
    }
  } else {
    // Currently ON → turn OFF only when below lower threshold (hysteresis)
    if (pctBW <= PCT_OFF) {  // below ~20% BW
      vibOn = false;
      aboveStart = 0;
    }
  }

  // 4) Apply to vibrator + LED (LED mirrors vibrator for easy visual feedback)
  digitalWrite(VIB_PIN, vibOn ? HIGH : LOW);
  digitalWrite(LED_PIN, vibOn ? HIGH : LOW);

  delay(50);  // small delay for serial readability
}
```